Unit 1

Artificial Intelligence -

Artificial Intelligence (AI) is a technology that enables machines and computers to perform tasks that typically require human intelligence. It allows systems to learn from data, recognise patterns, and make decisions to solve complex problems.

- Simulates human intelligence through learning and reasoning.
- Processes large amounts of data to identify patterns and insights.
- Continuously improves performance using experience and feedback.

Core Concepts

1. Machine Learning (ML)

Machine Learning is a subset of AI that enables systems to learn from data and improve performance without explicit programming.

- Identifies patterns and relationships from data.
- Improves predictions and decisions through experience.
- Powers applications like recommendation and prediction systems.

2. Generative AI

Generative AI focuses on creating new content such as text, images, audio, and videos using learned patterns from large datasets.

- Generates original content instead of only analyzing data.
- Learns patterns from massive datasets for content creation.
- Used in chatbots, image generation, and content writing tools.

3. Natural Language Processing (NLP)

Natural Language Processing (NLP) enables computers to understand, interpret, and generate human language naturally.

- Supports tasks like translation, sentiment analysis, and chatbots.
- Combines linguistics with computer science techniques.
- Enables voice assistants such as Siri and Alexa.

4. Expert Systems

Expert Systems simulate human decision-making using predefined rules and domain-specific knowledge.

- Uses if-then rules to solve specialized problems.

- Mimics reasoning of human experts in specific domains.
- Commonly applied in healthcare, finance, and diagnostics.

Working

1. **Data Collection**: AI systems rely on large sets of data which could include images, text or sensor readings. For example, teaching an AI to recognize cats, we collect a dataset of labeled cat images.
2. **Processing and Learning**: It uses algorithms to analyze data and identify patterns. For example, it learns to recognize key features like a cat's shape, ears or whiskers helping it understand the data.
3. **Model Training**: The AI model is trained using the data, adjusting its internal settings to improve its predictions. With more data, the model becomes more accurate and better at recognizing new examples like unseen images of cats.
4. **Decision Making**: Once trained, it can use what it has learned to make decisions. For example, it can find whether a new image contains a cat based on the patterns it learned during training.
5. **Feedback and Improvement**: It can improve through feedback, especially in methods like reinforcement learning. In this case, the AI receives rewards or penalties, refining its ability to make better decisions over time.

Types of Artificial Intelligence

AI can be classified into two main categories based on its capabilities and functionalities.

1. Based on Capabilities

- **Narrow AI**: Designed to perform specific tasks such as speech recognition or recommendation systems.
- **General AI**: A theoretical AI capable of performing human-like intellectual tasks across multiple domains.
- **Superintelligent AI**: A hypothetical AI that would surpass human intelligence in reasoning and decision-making.

2. Based on Functionalities

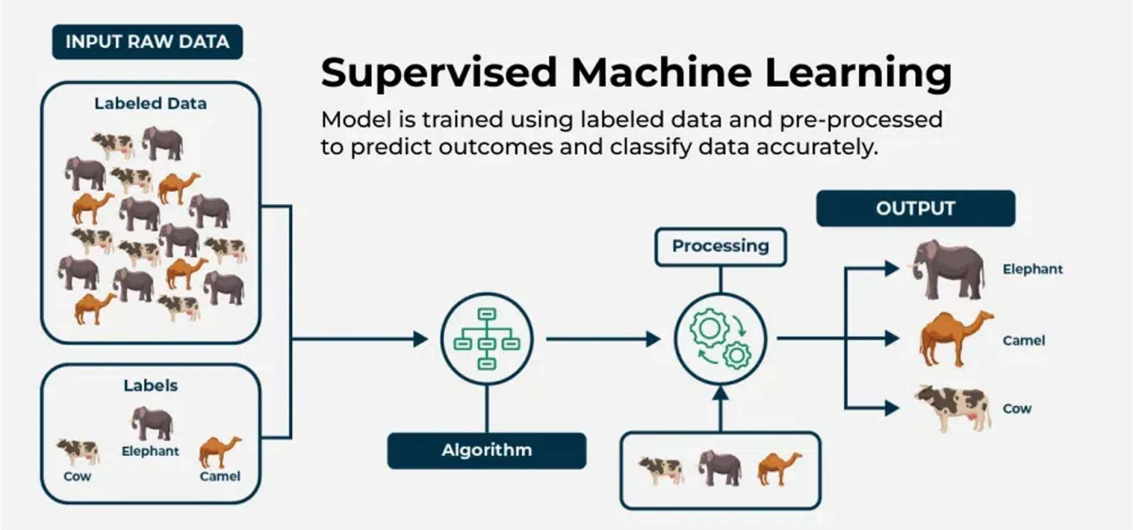
- **Reactive Machines**: AI systems that respond only to current inputs without storing past experiences.
- **Limited Memory**: AI systems that use past data and observations to improve future decisions.
- **Theory of Mind**: A theoretical AI that could understand human emotions, beliefs, and intentions.
- **Self-Aware AI**: A hypothetical AI with consciousness and awareness of its own existence.

AI Models- based on learning

AI models are systems that learn from data or predefined rules to make predictions and decisions. Different models use different learning approaches depending on the type of training data and feedback available. Based on Learning.

1. Supervised Learning Models

Supervised learning is a type of machine learning where a model learns from labelled data, meaning each input has a correct output. The model compares its predictions with actual results and improves over time to increase accuracy

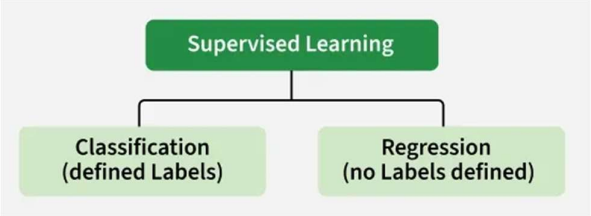


Its main features are:

- **Each input has a known output**
- **Adjusts itself to reduce prediction errors**
- **Make accurate predictions on new data**
- **For example it recognizing handwritten digits from trained data**

Types of Supervised Learning

Now, Supervised learning can be applied to two main types of problems:



- **Classification: Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).**
- **Regression: Where the output is a continuous variable (e.g., predicting house prices, stock prices).**

Let's first understand the classification and regression data through the table below:

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

Both the above figures have labelled data set as follows:

Figure A: It is a dataset of a shopping store that is useful in predicting whether a customer will purchase a particular product under consideration or not based on his/her gender, age and salary.

- **Input: Gender, Age, Salary**
- **Output: Purchased i.e. 0 or 1; 1 means yes the customer will purchase and 0 means that the customer won't purchase it.**

Figure B: It is a Meteorological dataset that serves the purpose of predicting wind speed based on different parameters.

- **Input: Dew Point, Temperature, Pressure, Relative Humidity, Wind Direction**
- **Output: Wind Speed**

Working of Supervised Machine Learning

The working of supervised machine learning follows these key steps:

1. Collect Labeled Data

- **Gather a dataset where each input has a known correct output (label).**
- **Example: Images of handwritten digits with their actual numbers as labels.**

2. Split the Dataset

- **Divide the data into training data (about 80%) and testing data (about 20%).**
- **The model will learn from the training data and be evaluated on the testing data.**

3. Train the Model

- **Feed the training data (inputs and their labels) to a suitable supervised learning algorithm (like Decision Trees, SVM or Linear Regression).**
- **The model tries to find patterns that map inputs to correct outputs.**

4. Validate and Test the Model

- **Evaluate the model using testing data it has never seen before.**
- **The model predicts outputs and these predictions are compared with the actual labels to calculate accuracy or error.**

5. Deploy and Predict on New Data

- **Once the model performs well, it can be used to predict outputs for completely new, unseen data.**

Supervised Machine Learning Algorithms

Supervised learning includes different types of algorithms used to predict outputs based on labeled data. Each algorithm is designed for specific tasks like prediction or classification.

- **Linear Regression: Used to predict continuous values (e.g., price, temperature). It is simple and widely used.**
- **Logistic Regression: Logistic regression is a type of supervised learning classification algorithm that is used to predict a binary output variable.**
- **Decision Trees: Uses a tree-like structure where each node represents a decision and each leaf represents an outcome.**
- **Random Forests: Combines multiple decision trees to improve accuracy and reduce overfitting.**
- **Support Vector Machine(SVM): Separates data into classes using a boundary (hyperplane). Support vectors help define this boundary.**
- **K-Nearest Neighbors: Predicts based on the closest data points. Results depend on the value of k and distance measure.**
- **Gradient Boosting: Builds models step-by-step by correcting errors of previous models, creating a strong model.**
- **Naive Bayes Algorithm: Based on probability and Bayes' Theorem, assuming features are independent of each other.**

Examples

- **Fraud Detection in Banking: Uses labeled transaction data to identify and predict fraudulent activities.**
- **Parkinson Disease Prediction: Analyzes medical data such as voice recordings, motor symptoms, and clinical measurements to identify patterns and predict whether a patient has disease or not.**

- **Customer Churn Prediction: Uses historical customer data to predict whether a customer will leave a service.**
- **Cancer cell classification: Implements supervised learning for cancer cells based on their features and identifying them if they are 'malignant' or 'benign.**
- **Stock Price Prediction: Uses past data to predict stock trends and support investment decisions.**

Advantages

- **Easy to understand and implement as it learns from labeled data.**
- **Provides high accuracy when sufficient labeled data is available.**
- **Works for both classification (spam detection, disease prediction) and regression (price forecasting).**
- **Can generalize well to unseen data with proper training and diverse datasets.**
- **Widely used in applications like speech recognition, medical diagnosis, sentiment analysis and fraud detection.**

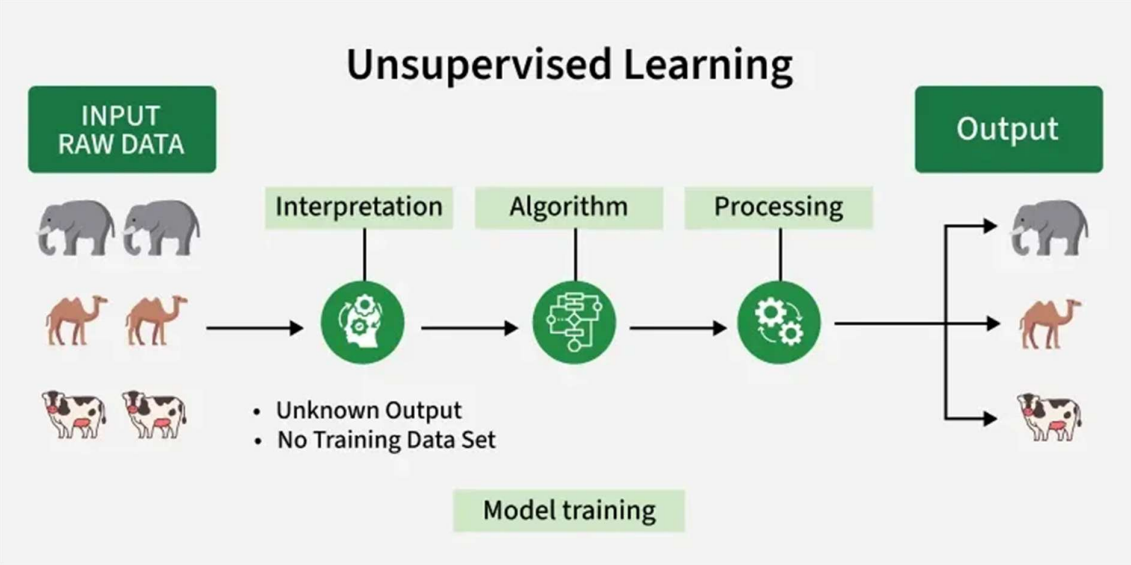
Disadvantages

- **Requires large amounts of labeled data, which is expensive and time-consuming to prepare.**
- **Can be biased if the training data is unbalanced, leading to unfair or inaccurate predictions.**
- **May overfit the training data instead of learning general patterns, especially with small datasets.**
- **Performance can drop when applied to data that is very different from the training data.**
- **Not easily scalable for problems with a very large number of labels, such as in natural language tasks.**

2. Unsupervised Learning Models

Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.

- **It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.**
- **Helps identify hidden patterns in data**
- **Useful for grouping, compression and anomaly detection**



The image shows set of animals like elephants, camels and cows that represents raw data that the unsupervised learning algorithm will process.

- **The "Interpretation" stage signifies that the algorithm doesn't have predefined labels or categories for the data. It groups the data based on inherent patterns.**
- **An algorithm represents unsupervised learning process that helps to identify patterns in the data.**
- **The processing stage shows the algorithm working on the data.**

The output shows the results of the unsupervised learning process. In this case, the algorithm might have grouped the animals into clusters based on their species (elephants, camels, cows).

Working of Unsupervised Learning

The working of unsupervised machine learning can be explained in these steps:

1. Collect Unlabeled Data

- **Gather a dataset without predefined labels or categories.**
- **Example: Images of various animals without any tags.**

2. Select an Algorithm

- **Choose a suitable unsupervised algorithm such as clustering like K-Means, association rule learning like Apriori or dimensionality reduction like PCA based on the goal.**

3. Train the Model on Raw Data

- **Feed the entire unlabeled dataset to the algorithm.**

- **The algorithm looks for similarities, relationships or hidden structures within the data.**

4. Group or Transform Data

- **The algorithm organizes data into groups (clusters), rules or lower-dimensional forms without human input.**
- **Example: It may group similar animals together or extract key patterns from large datasets.**

5. Interpret and Use Results

- **Analyze the discovered groups, rules or features to gain insights or use them for further tasks like visualization, anomaly detection or as input for other models.**

Unsupervised Learning Algorithms

There are mainly 3 types of Unsupervised Algorithms that are used:

1. Clustering Algorithms

Clustering is an unsupervised machine learning technique that groups unlabeled data into clusters based on similarity. Its goal is to discover patterns or relationships within the data without any prior knowledge of categories or labels.

- **Groups data points that share similar features or characteristics.**
- **Helps find natural groupings in raw, unclassified data.**
- **Commonly used for customer segmentation, anomaly detection and data organization.**
- **Works purely from the input data without any output labels.**
- **Enables understanding of data structure for further analysis or decision-making.**

2. Association Rule Learning

Association rule learning is a rule-based unsupervised learning technique used to discover interesting relationships between variables in large datasets. It identifies patterns in the form of "if-then" rules, showing how the presence of some items in the data implies the presence of others.

- **Finds frequent item combinations and the rules connecting them.**
- **Commonly used in market basket analysis to understand product purchase relationships.**
- **Helps retailers design promotions and cross-selling strategies.**

Some common Association Rule Learning algorithms:

- ***Apriori Algorithm: Finds patterns by exploring frequent item combinations step-by-step.***
- ***FP-Growth Algorithm: An Efficient Alternative to Apriori. It quickly identifies frequent patterns without generating candidate sets.***
- ***Eclat Algorithm: Uses intersections of itemsets to efficiently find frequent patterns.***

- ***Efficient Tree-based Algorithms**: Scales to handle large datasets by organizing data in tree structures.*

3. Dimensionality Reduction

Dimensionality reduction is the process of decreasing the number of features or variables in a dataset while retaining as much of the original information as possible. This technique helps simplify complex data making it easier to analyze and visualize. It also improves the efficiency and performance of machine learning algorithms by reducing noise and computational cost.

- **It reduces the dataset's feature space from many dimensions to fewer, more meaningful ones.**
- **Helps focus on the most important traits or patterns in the data.**
- **Commonly used to improve model speed and reduce overfitting.**

Here are some popular Dimensionality Reduction algorithms:

- ***Principal Component Analysis (PCA)**: Reduces dimensions by transforming data into uncorrelated principal components.*
- ***Non-negative Matrix Factorization (NMF)**: Breaks data into non-negative parts to simplify representation.*
- ***Locally Linear Embedding (LLE)**: Reduces dimensions while preserving the relationships between nearby points.*
- ***Isomap**: Captures global data structure by preserving distances along a manifold.*

Applications

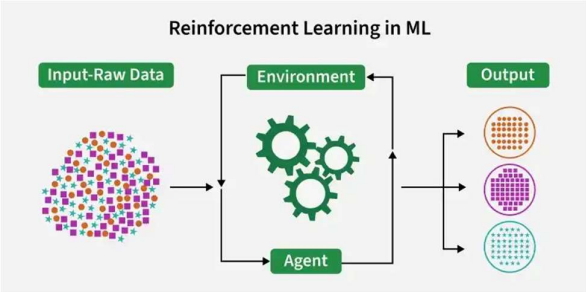
- **Customer Segmentation: Algorithms cluster customers based on purchasing behavior or demographics, enabling targeted marketing strategies.**
- **Anomaly Detection: Identifies unusual patterns in data, aiding fraud detection, cybersecurity and equipment failure prevention.**
- **Recommendation Systems: Suggests products, movies or music by analyzing user behavior and preferences.**
- **Image and Text Clustering: Groups similar images or documents for tasks like organization and content recommendation.**
- **Social Network Analysis: Detects communities or trends in user interactions on social media platforms.**

Advantages

- **Works with raw, unlabeled data, saving time and effort required for data annotation.**
- **Finds hidden patterns and natural groupings in data that may not be easily identified by humans.**
- **Handles large and complex datasets efficiently, including high-dimensional data.**
- **Helps detect anomalies and unusual data points without needing prior examples.**

3. Reinforcement Learning Mode.

Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards. RL allows machines to learn by interacting with an environment and receiving feedback based on their actions. This feedback comes in the form of rewards or penalties.



Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal. The agent performs actions and receives feedback to optimize its decision-making over time.

- **Agent**: The decision-maker that performs actions.
- **Environment**: The world or system in which the agent operates.
- **State**: The situation or condition the agent is currently in.
- **Action**: The possible moves or decisions the agent can make.
- **Reward**: The feedback or result from the environment based on the agent's action.

Types of Reinforcements

1. Positive Reinforcement: Positive Reinforcement is defined as when an event, occurs due to a particular behavior, increases the strength and the frequency of the behavior. In other words, it has a positive effect on behavior.

- **Advantages**: Maximizes performance, helps sustain change over time.
- **Disadvantages**: Overuse can lead to excess states that may reduce effectiveness.

2. Negative Reinforcement: Negative Reinforcement is defined as (strengthening) rise of behavior because a negative condition is stopped or avoided.

- **Advantages**: Increases behavior frequency, ensures a minimum performance standard.
- **Disadvantages**: It may only encourage just enough action to avoid penalties.

Application

- **Robotics**: RL is used to automate tasks in structured environments such as manufacturing, where robots learn to optimize movements and improve efficiency.
- **Games**: Advanced RL algorithms have been used to develop strategies for complex games like chess, Go and video games, outperforming human players in many instances.
- **Industrial Control**: RL helps in real-time adjustments and optimization of industrial operations, such as refining processes in the oil and gas industry.
- **Personalized Training Systems**: RL enables the customization of instructional content based on an individual's learning patterns, improving engagement and effectiveness.